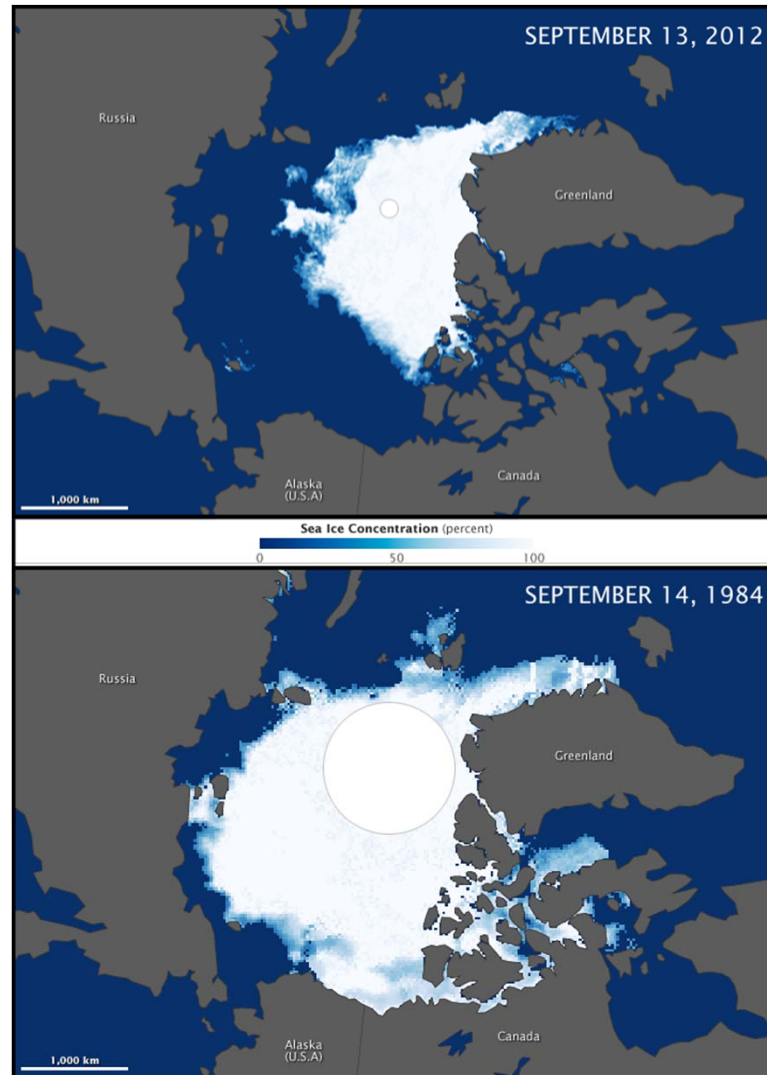


Climate Change Risk and Adaptation: a Greater Manchester perspective

Dr Jeremy Carter – University of Manchester

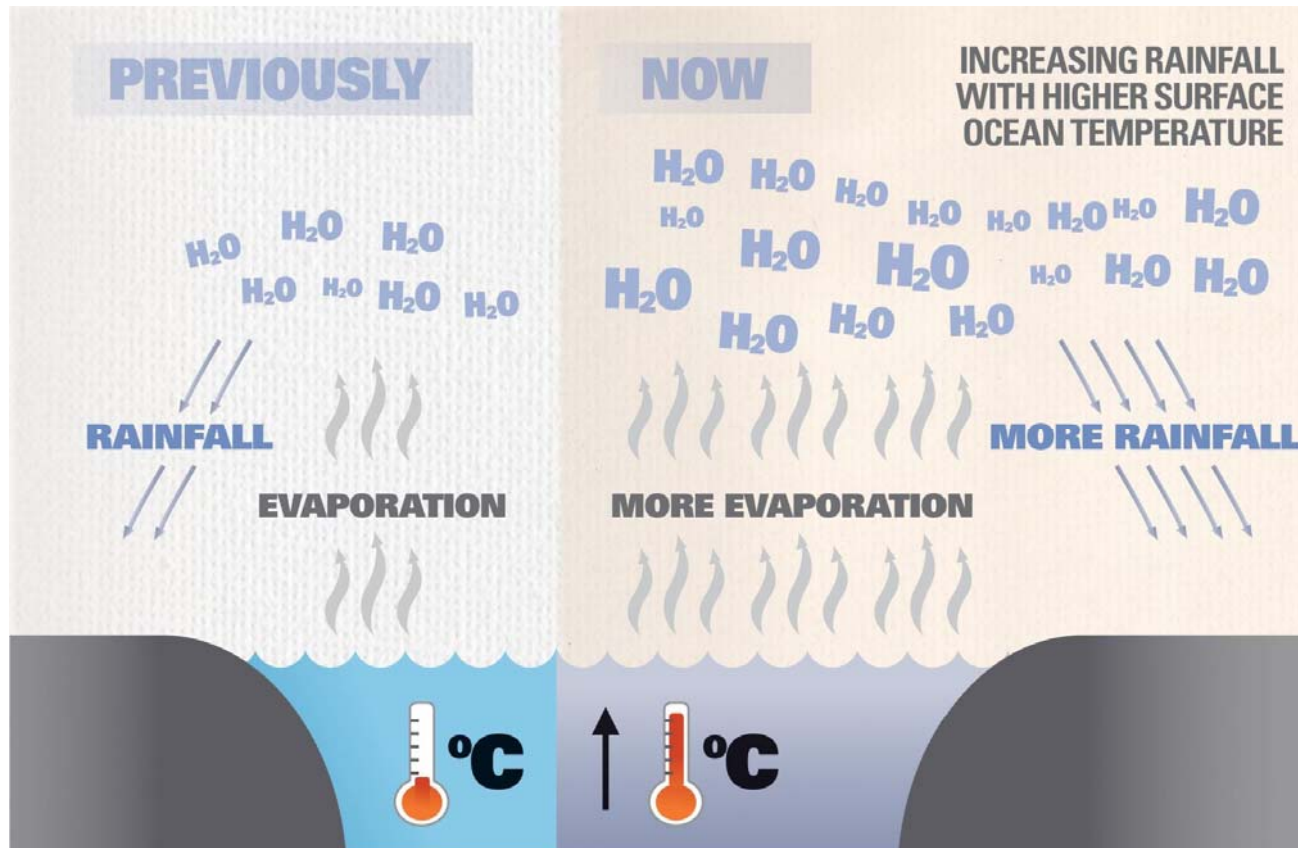
**APSE Renewables and Climate Change Advisory
Group (14-12-13)**

The Adaptation Imperative



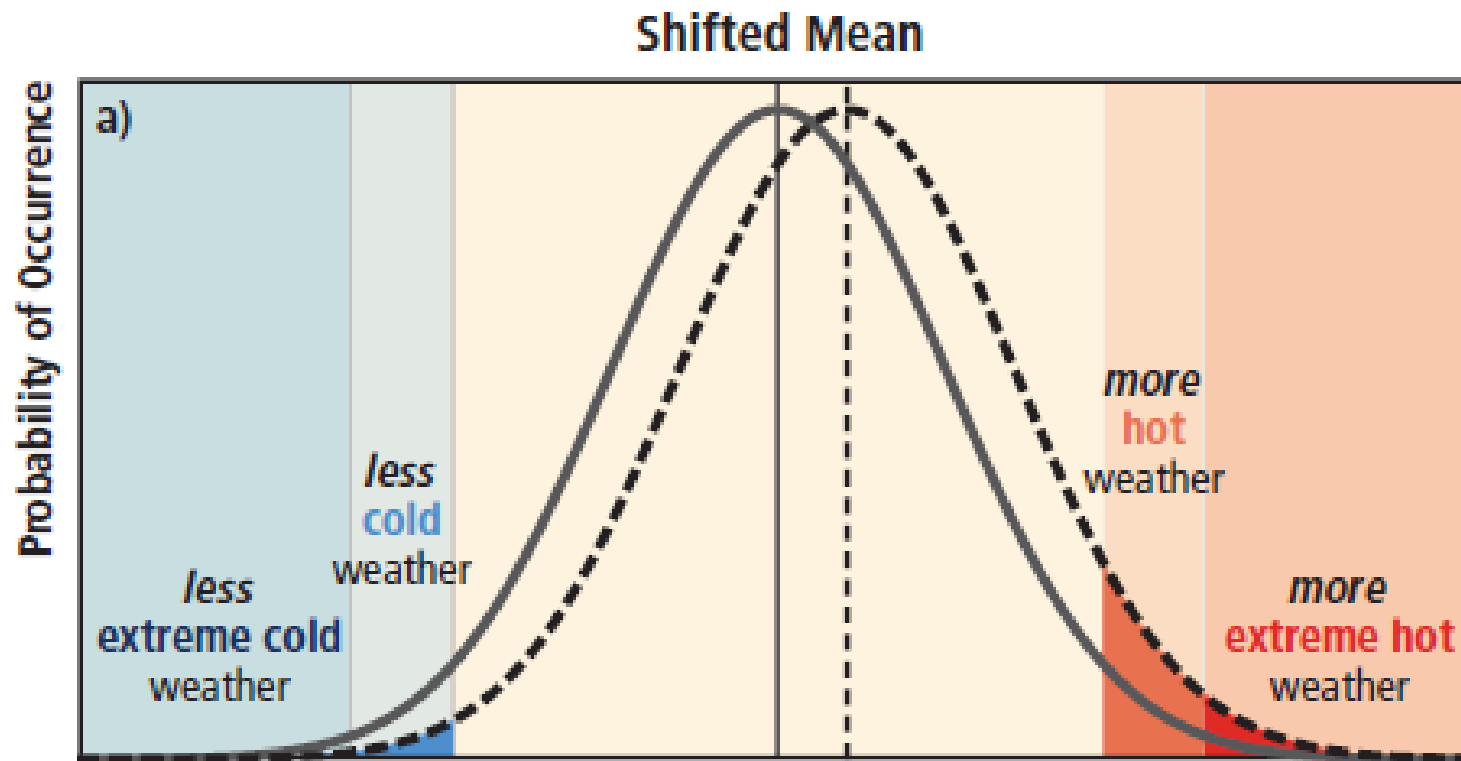
Climate change today

- *‘... human-induced climate change is not an issue for the next decades or century: it is an issue facing us today.’ (Bob Watson 2011)*



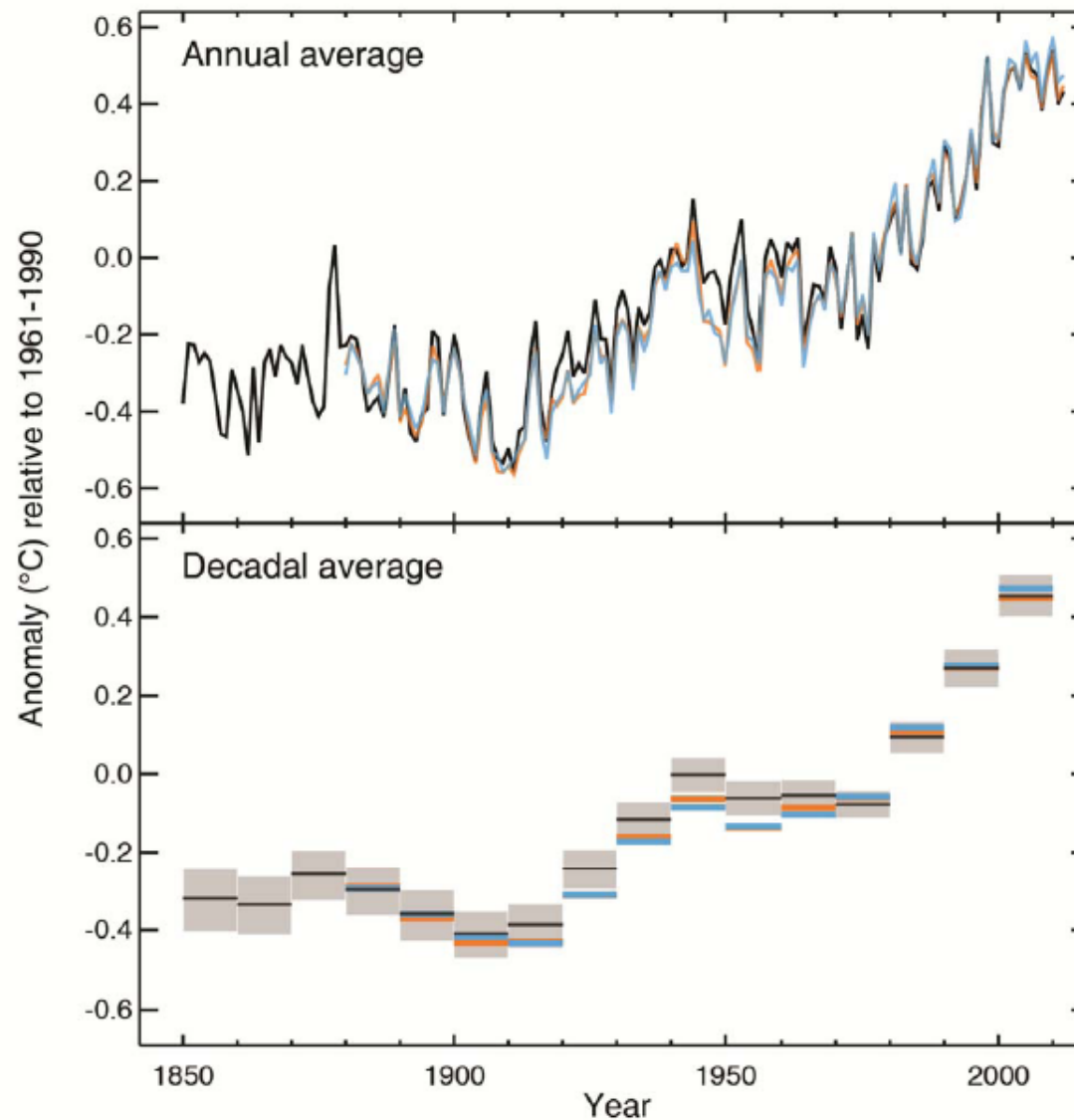
Source: Australian
Bureau of
Meteorology
(2013)

Effect of shift in temperature distribution towards a warmer climate



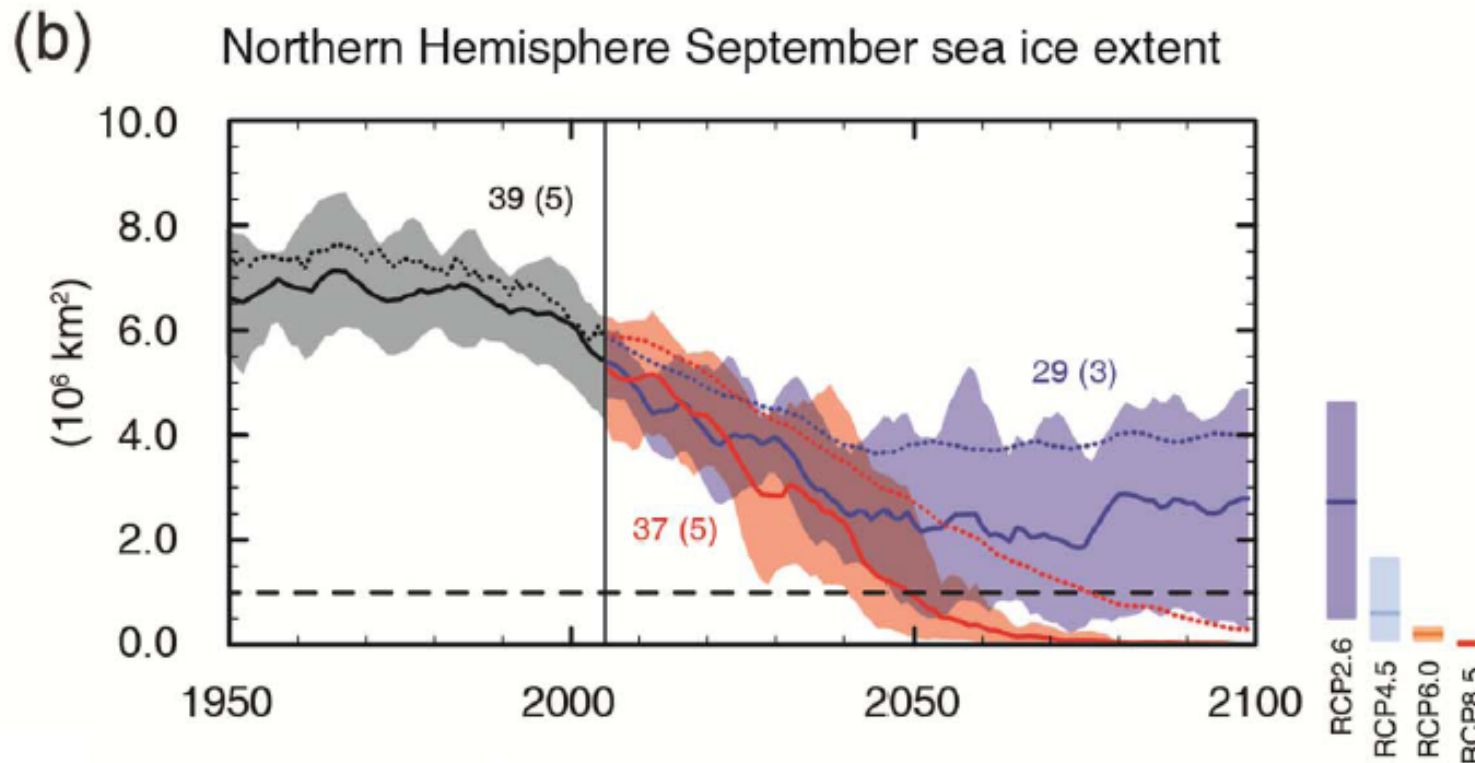
Source: IPCC 2012

Observed globally averaged combined land and ocean surface temperature anomaly (1850-2012)



Source: IPCC AR5 (2013)

Northern Hemisphere September sea ice extent – observations and projections

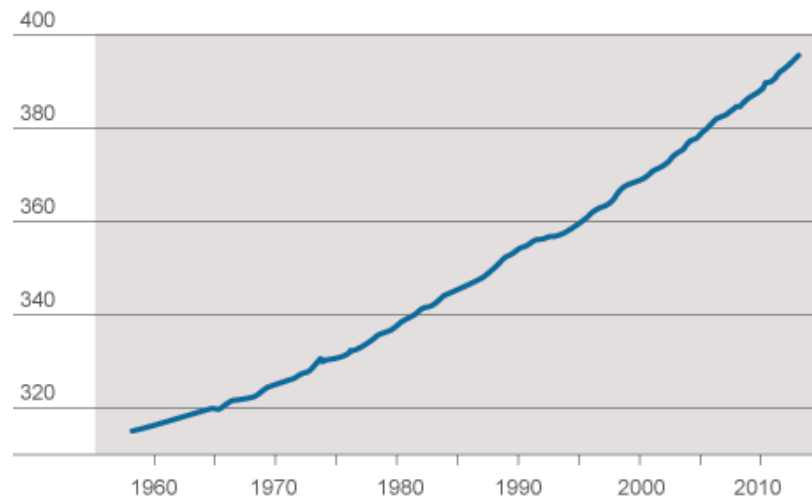


Source: IPCC AR5 (2013)

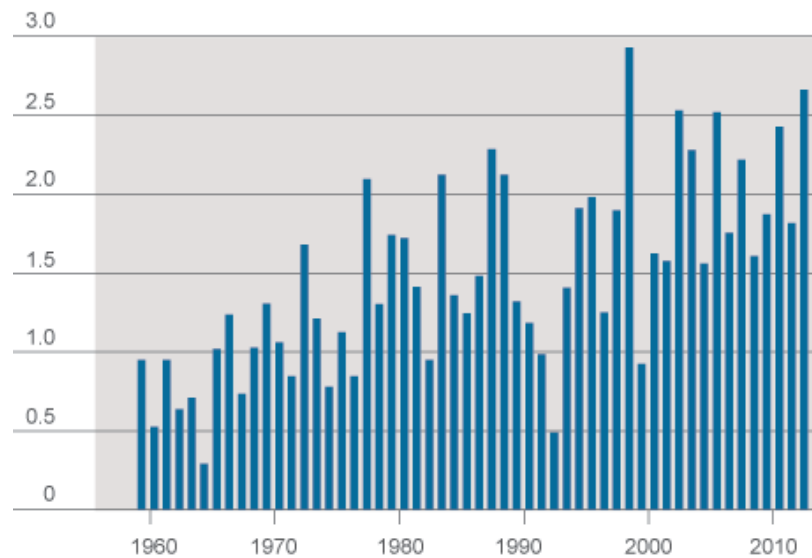


Atmospheric carbon increases

Parts per million, at Mauna Loa observator, seasonally adjusted



Annual mean growth rate, parts per million per year



SOURCE: NOAA, SCRIPPS INSTITUTE OF OCEANOGRAPHY

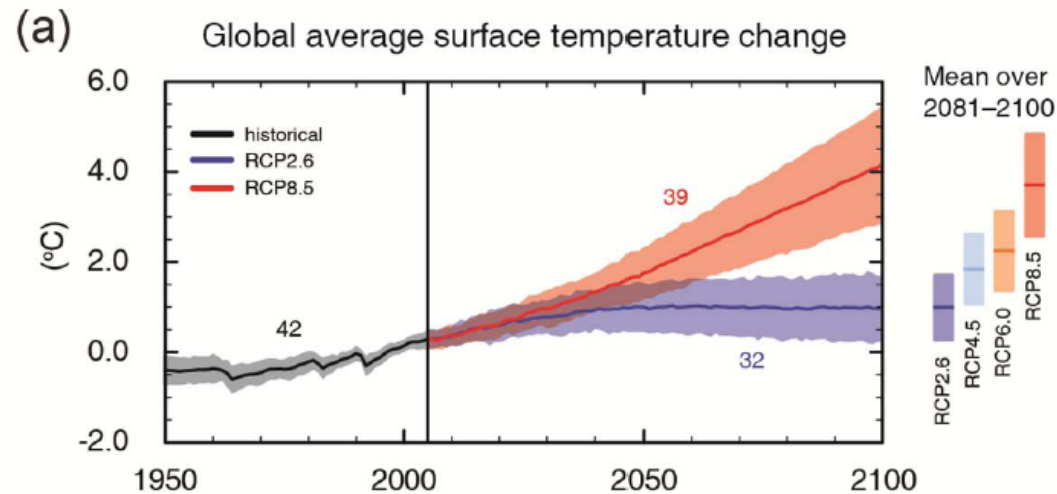
Carbon Emissions

- Atmospheric CO² emissions have crossed the 400 ppm mark
- 2012 saw the 2nd highest rise in global CO² emissions

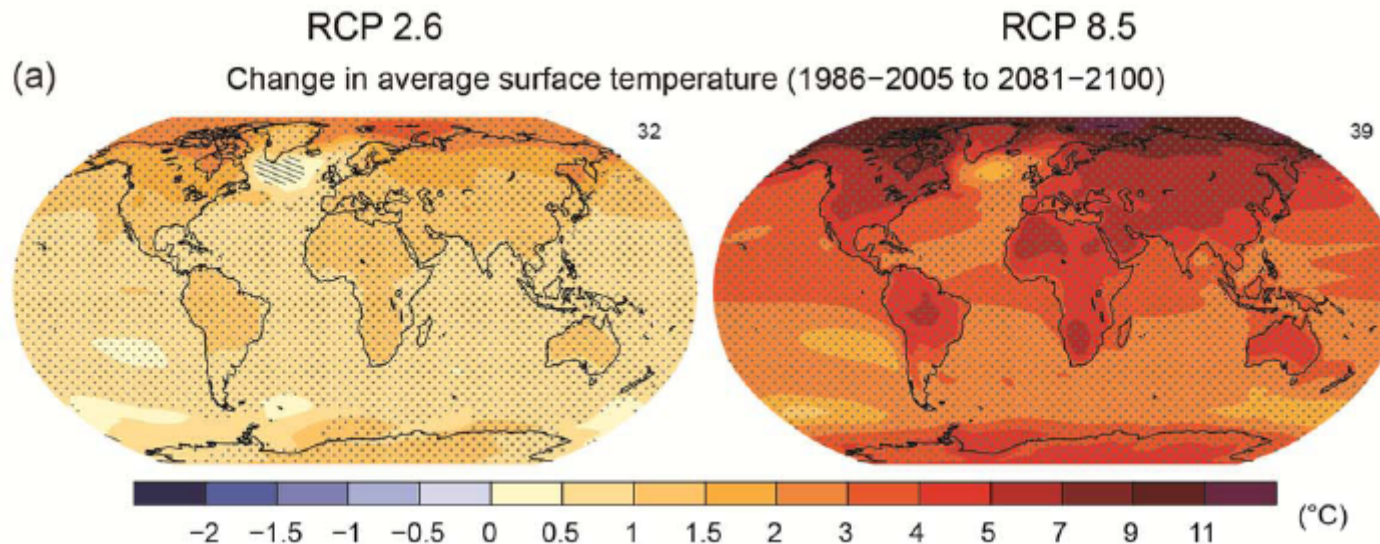
Key factors influencing future greenhouse gas emissions

Factor affecting GHG emissions	Trend
Global economic activity (via the use of fossil fuels and land use change).	Increasing
The carbon intensity of the economy.	Declining
The functioning of unmanaged land and ocean based carbon sources and sinks.	Declining

Global surface temperature change and projections



“At best we have a 50/50 chance of a 3°C world and 5°C is not out of the question” *Bob Watson (Former Defra Chief Scientific Advisor)*.



Source: IPCC AR5 (2013)

Climate Change Risk and Adaptation in Greater Manchester



Recent trends in GM weather and climate hazards

Weather/ climate event	Number of recorded events across GM (1945-2008)	% of total recorded events across GM (1945-2008)
Flood	158	41.9
Storm	85	22.5
Cold	63	16.7
Fog	28	7.4
Heat	22	5.8
Drought	8	2.1
Air Quality	7	1.9
Smog	6	1.7

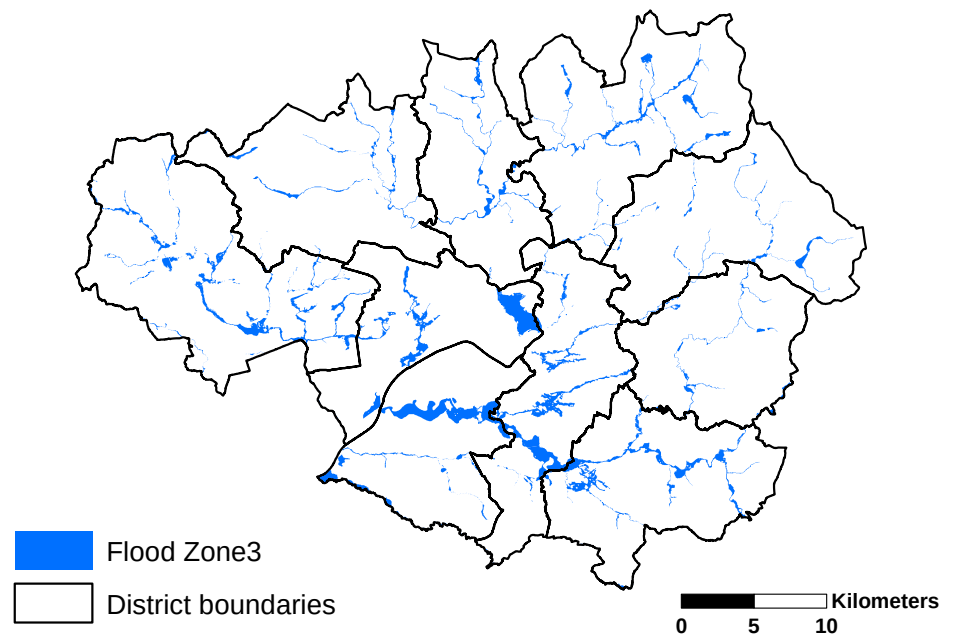
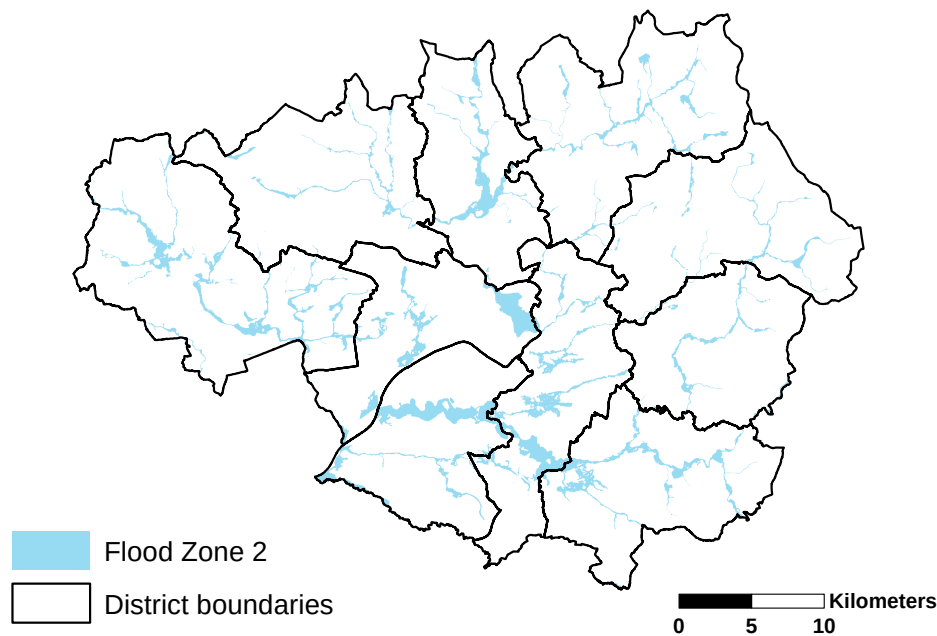
Projected changes to GM's climate (2050s high emissions scenario for the Mersey Basin)

Variable	Baseline (1961-1990)	2050s - 50 th percentile projection	2050s - 90 th percentile projection
Annual mean temperature (°C)	9-10 °C	+2.4 °C	+3.6 °C
Change in warmest day in summer (°C)	26-27 °C	+3.1 °C	+6 °C
Summer mean precipitation	200-250mm	-20%	-36%
Winter mean precipitation	200-250mm	+14%	+28%

(Source: Cavan 2010)

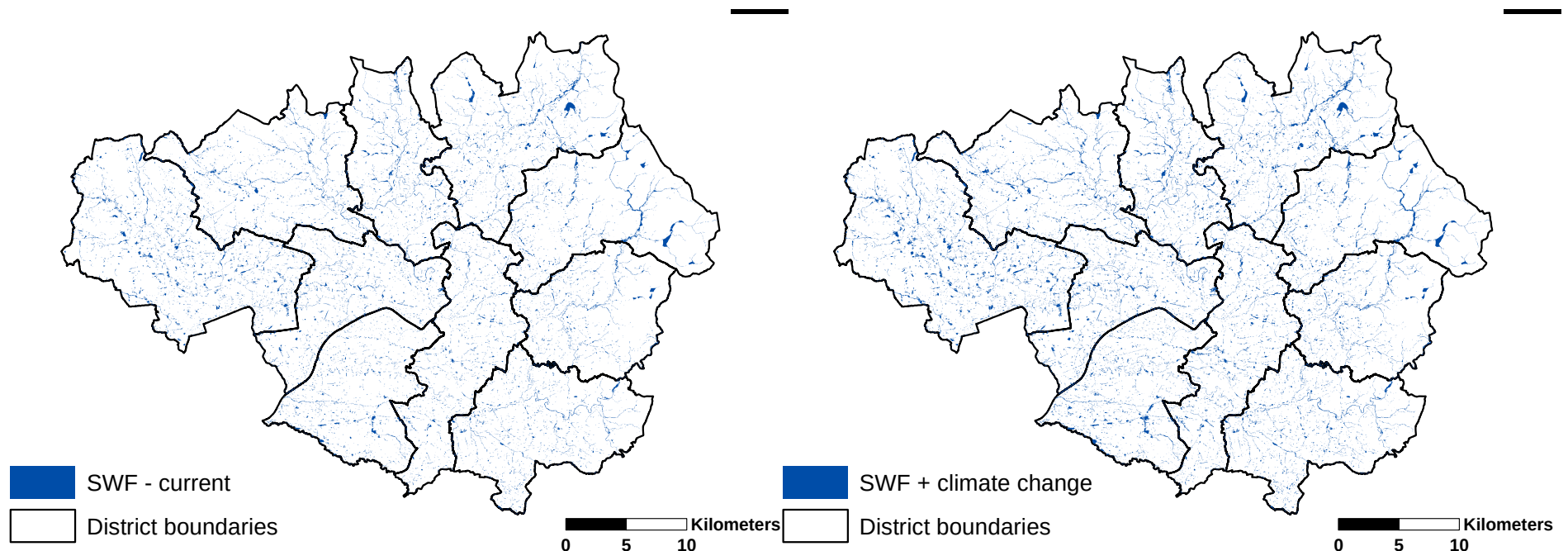
River flooding

- Flood Zone 2: 1 in 100 to 1 in 1000 years - 6.8% GM area
- Flood Zone 3: 1 in 100 years or greater – 4.6% GM area

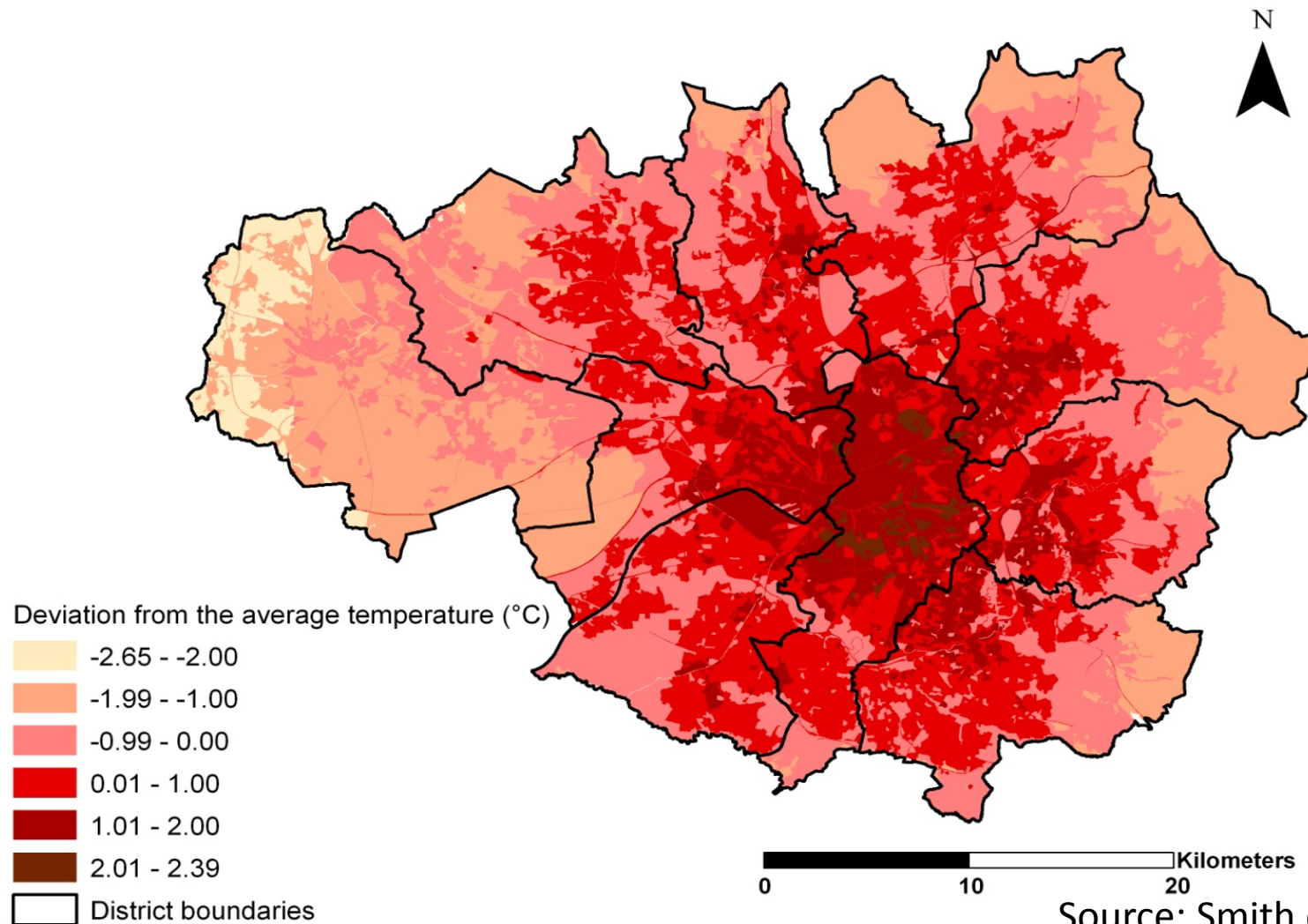


Surface Water Flooding (SWF)

- 1 in 200 year event; 30cm flood depth
 - 2.8% of GM area currently at risk
 - 3.8% at risk in the future with climate change



Greater Manchester's Urban Heat Island Effect



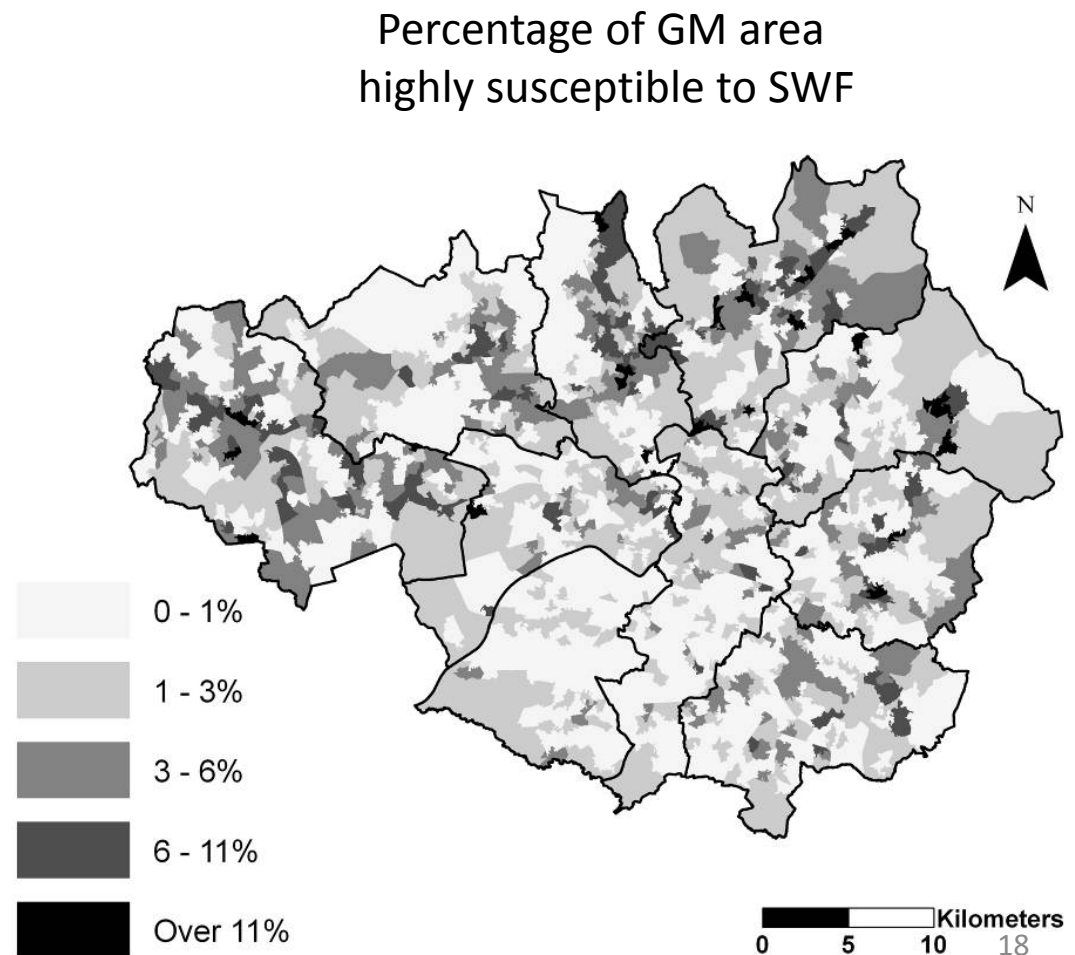
Source: Smith et al., 2009

Vulnerability to climate change

- Vulnerability is a state; a starting point from which hazards are experienced.
- Associated with people, places or things.
- Influenced by factors underlying potential ***exposure*** and ***sensitivity*** to climate hazards.

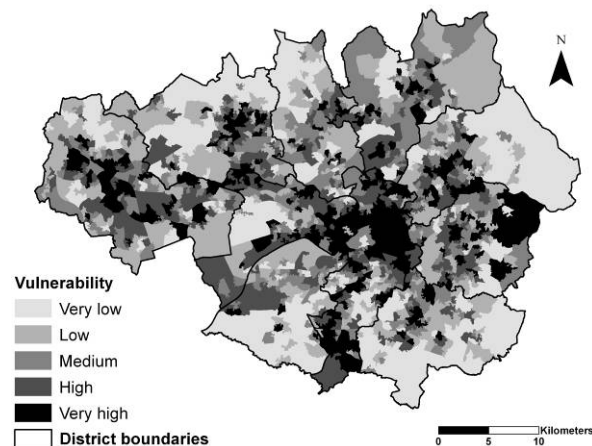
Exposure to surface flooding in Greater Manchester

2.2% of Greater Manchester is highly susceptible to surface water flooding (depth of 1m)

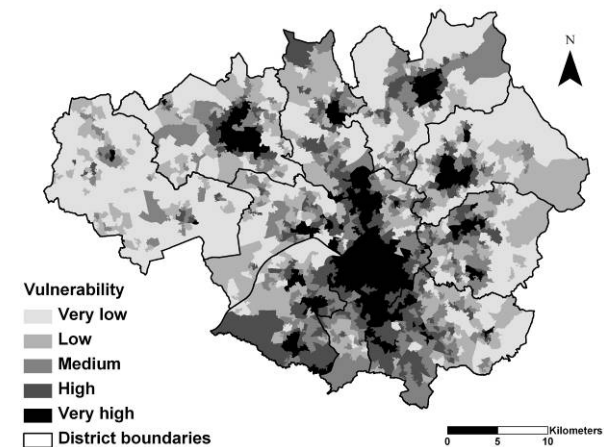


Sensitivity of people to climate change hazards

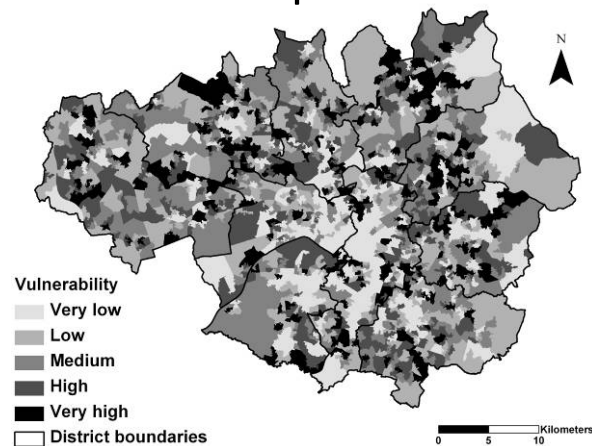
Poverty and poor health



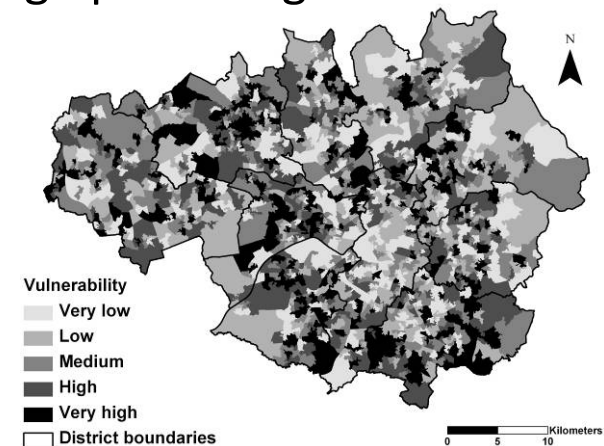
Diverse communities



Families with dependent children

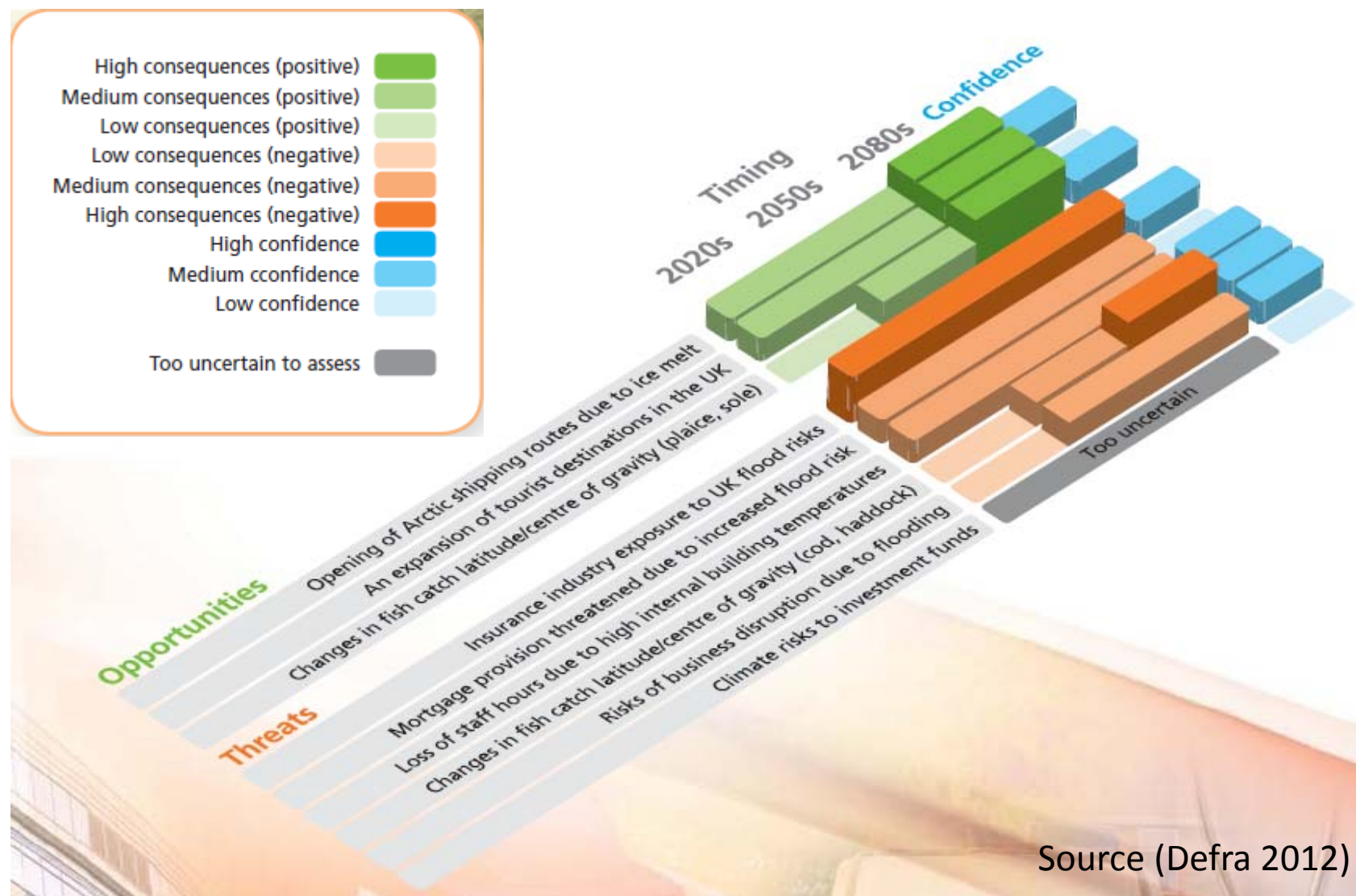


High percentage of the elderly



UK Climate Change Risk Assessment 2012

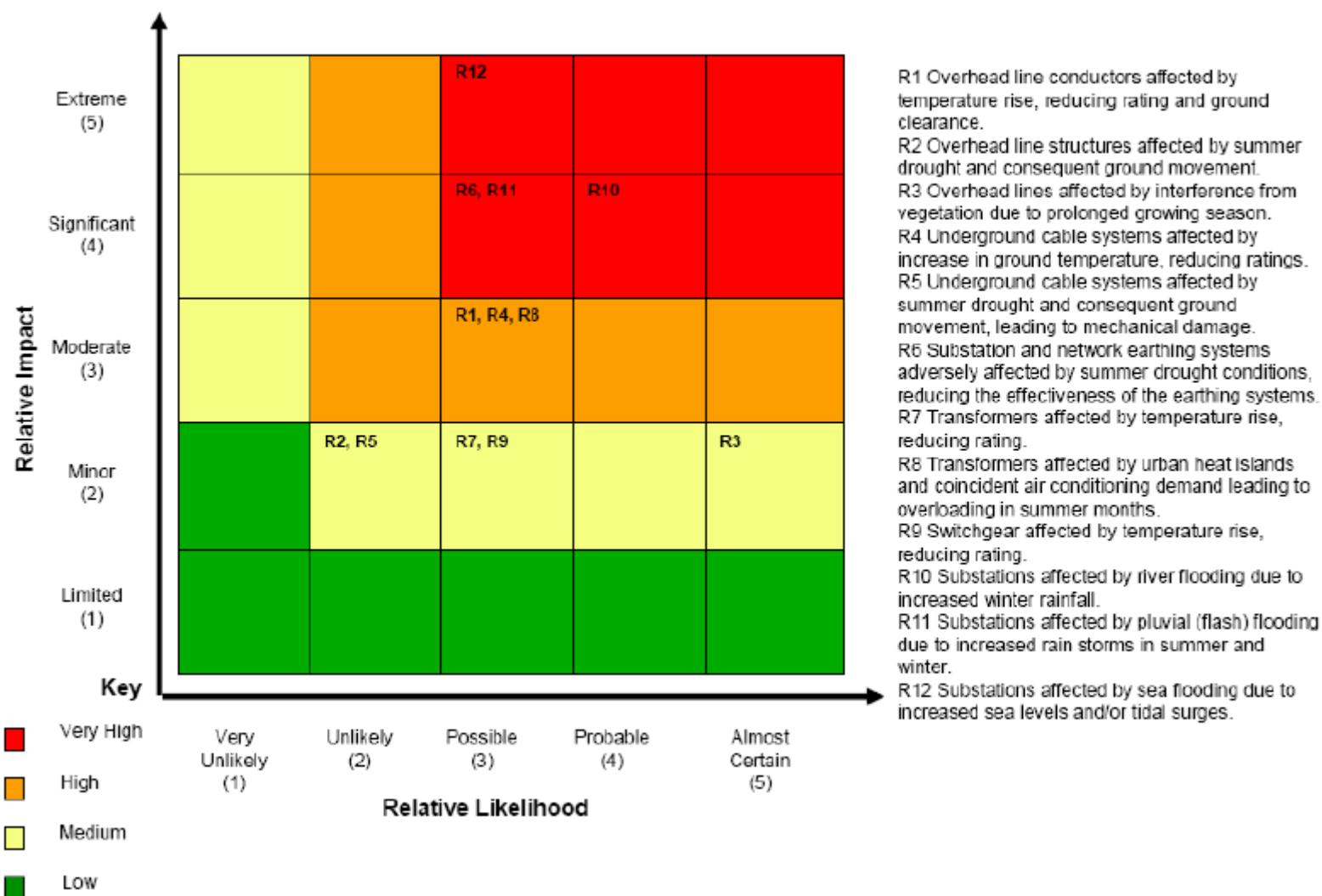
(Impacts on Business)



Source (Defra 2012)

Risk assessment output example: Electricity North West

ENA Adaptation to Climate Change Risk Matrix (if no adaptation measures taken)



Climate Change Risks to the Greater Manchester Strategy (GMS): a spatial approach

Six GMS themes assessed

- Housing development areas
- Regional centre and town centres
- Employment locations
- Transport infrastructure
- Science and innovation assets
- Energy and water infrastructure

Three weather/climate hazards

- River Flooding (based on EA flooding maps)
- Surface water flooding (based on JBA Consulting model results)
- High Temperatures (based on UoM heat island mapping)

Risk Assessment Method

Likelihood of occurrence of climate change impact	High (3)	3	6	9
	Medium (2)	2	4	6
	Low (1)	1	2	3
		Low (1)	Medium (2)	High (3)
		Magnitude of consequence for GMS goals		

Likelihood = chance of a hazard event occurring x level of potential exposure to the hazard

Consequence of 13 different climate change impacts assessed according to whether they could negatively affect themes linked to the achievement of GMS priorities.

Likelihood of weather and climate hazards affecting GM's regional centre and different town centres.

Regional centre and town centres									
	Regional Centre	Altrincham	Ashton	Bolton	Bury	Oldham	Rochdale	Stockport	Wigan
FZ3	High likelihood	Low Likelihood – no potential exposure	Medium Likelihood	Medium Likelihood	Medium Likelihood	Low Likelihood – no potential exposure	High likelihood	High likelihood	High likelihood
FZ2	High likelihood	Low Likelihood – no potential exposure	Medium Likelihood	Medium Likelihood	Medium Likelihood	Low Likelihood – no potential exposure	High likelihood	High likelihood	High likelihood
SWF	Medium Likelihood	Medium Likelihood	Medium Likelihood	High likelihood	Medium Likelihood	Medium Likelihood	High likelihood	Medium Likelihood	High likelihood
SWF+cc	Medium Likelihood	Medium Likelihood	Medium Likelihood	High likelihood	Medium Likelihood	Medium Likelihood	High likelihood	Medium Likelihood	High likelihood
Heat now	Medium Likelihood	Medium Likelihood	Medium Likelihood	Low Likelihood	Medium Likelihood	Medium Likelihood	Low Likelihood	Medium Likelihood	Low Likelihood
Heat+cc	High likelihood	High likelihood	High likelihood	Low Likelihood	High likelihood	High likelihood	Low Likelihood	High likelihood	Low Likelihood
KEY	High likelihood		Medium Likelihood			Low Likelihood		Low Likelihood – no potential exposure	

Risk of weather and climate change impacts to GM's regional centre and town centres

		Regional Centre	Altrincham	Ashton	Bolton	Bury	Oldham	Rochdale	Stockport	Wigan
I1	FZ2+3									
	SWF									
I2	FZ2+3									
	SWF									
I4	FZ2+3									
	SWF									
I5	HW									
	HW+cc									
I6	HW									
	HW+cc									
I7	HW									
	HW+cc									
KEY										
I1 – Flood damage to buildings (residential, retail and commercial, public) I2 – Increased pressure on emergency services during flood events I4 – Higher insurance premiums in areas prone to flooding I5 – Reduction in worker productivity during heat waves I6 – Health problems during heat wave events I7 – Increased energy use during heat wave events							FZ2+3: Flood Zones 2 and 3 SWF: Surface Water Flooding HW: Heat Wave HW+cc: Heat Wave with Climate Change			
High risk			Medium risk		Low risk			Low risk - no potential exposure		

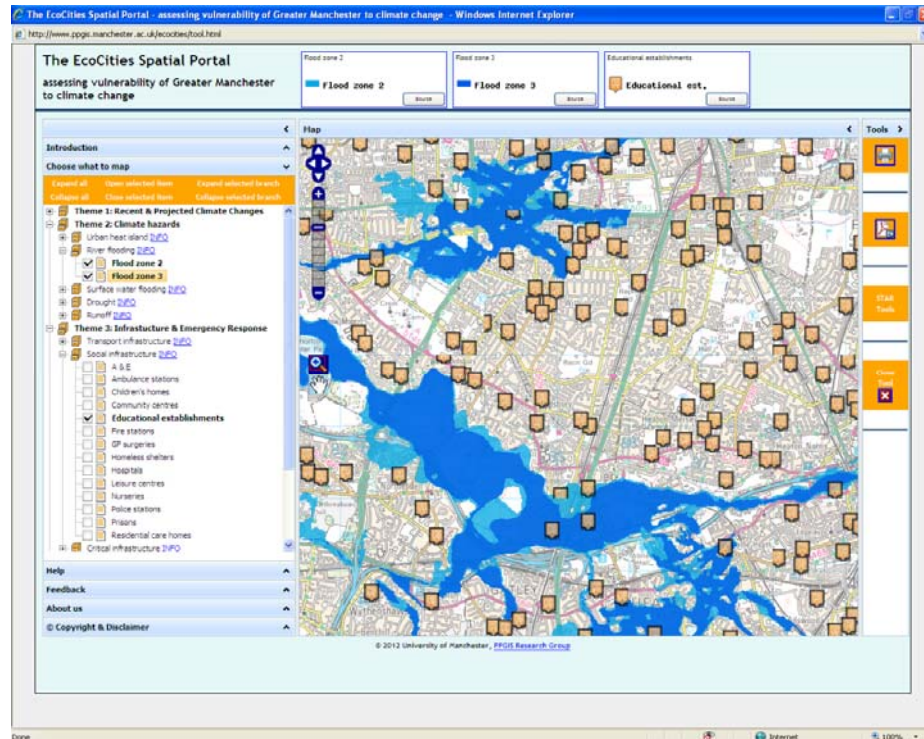
Outcome – risk prioritisation

Red risks: Active management to reduce risks through emergency planning and long term adaptation.

Amber risks: Contingency planning and frequent review to monitor underlying conditions. Consider adaptation response.

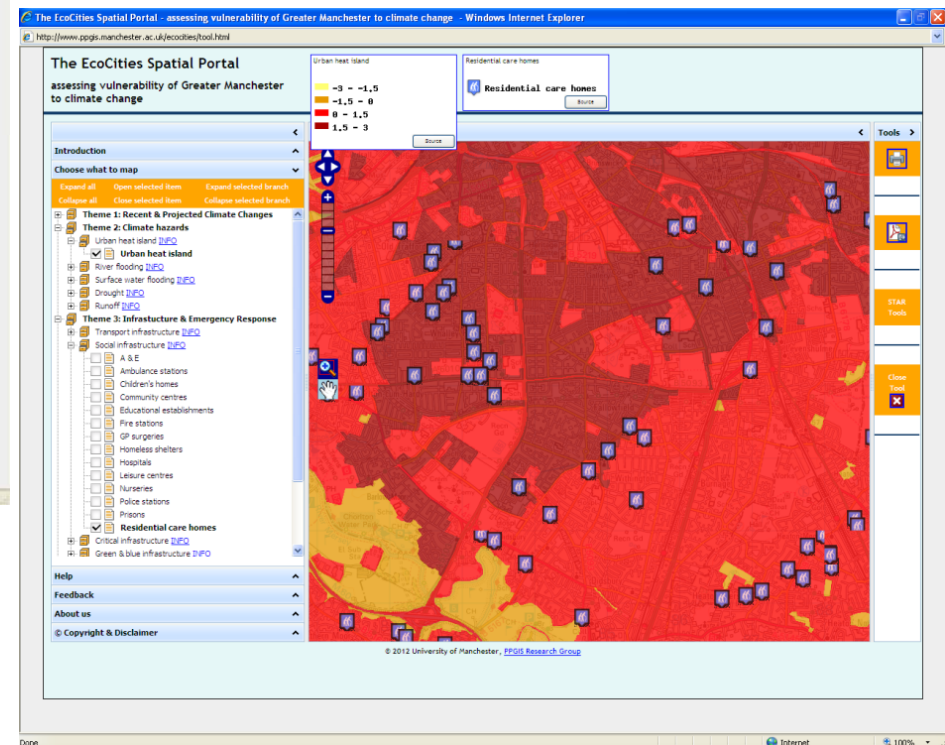
Green risks: Review risks periodically to ensure that conditions have not changed.

EcoCities spatial portal



Mapping flood zones 2 and 3, and educational establishments.

www.adaptingmanchester.co.uk



Mapping the urban heat island and residential care homes.

The EcoCities Spatial Portal

assessing the geography of
Greater Manchester's
vulnerability to climate change

Flood depth over 1m

Map Transparency: 75 %

- 0 - 1
- 1 - 3
- 3 - 6
- 6 - 11
- 11 - 21

Source

Educational establishments

Map Transparency: 75 %



Educational est.

Source

Districts

Map Transparency: 75 %

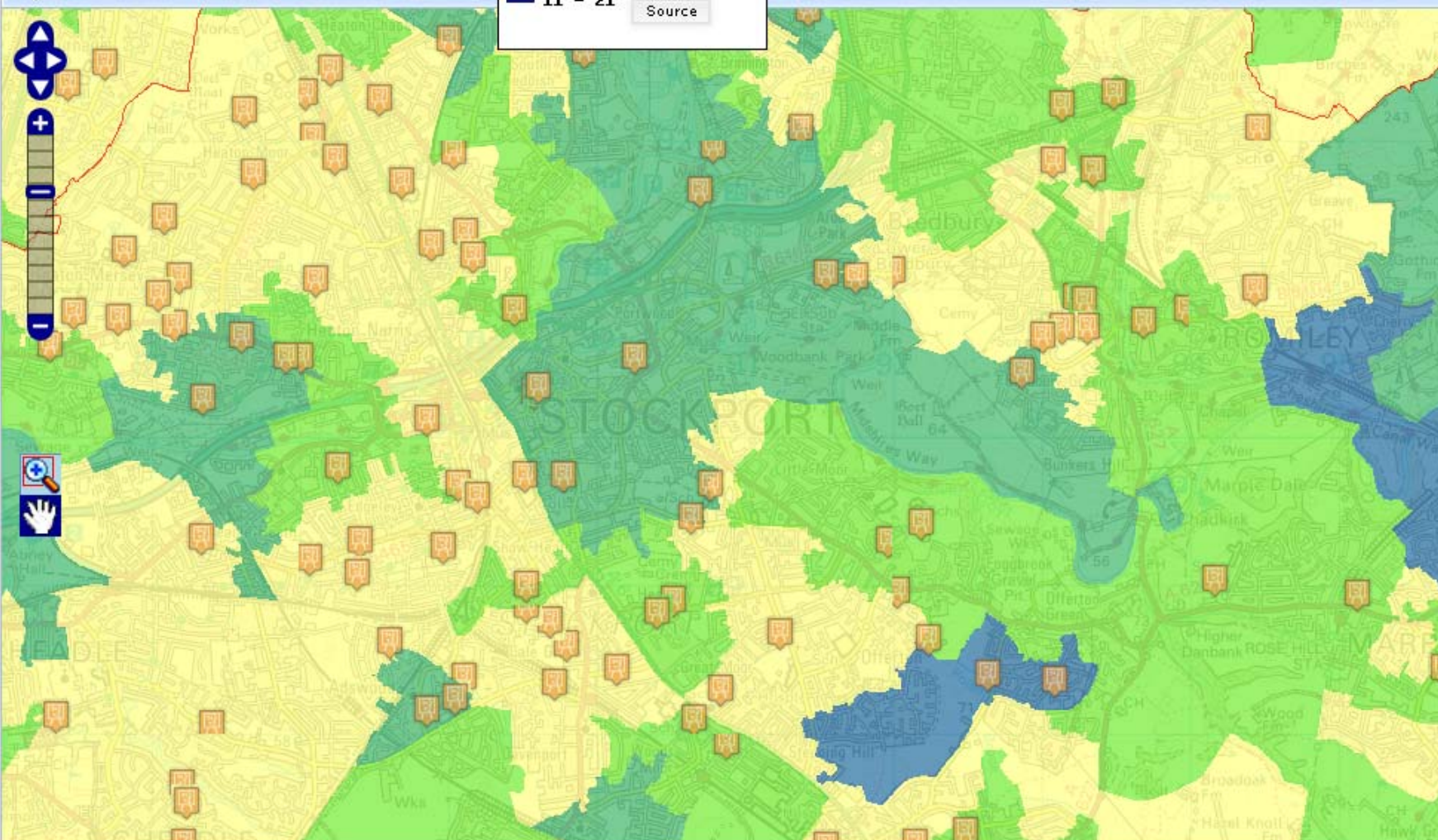


GM districts

Source

Map

Map Layer Menu



The EcoCities Spatial Portal

assessing the geography of
Greater Manchester's
vulnerability to climate change

Urban heat island

Map Transparency: 75 %

-3 - -1.5°C

-1.5 - 0°C

0 - 1.5°C

1.5 - 3°C

Source

Residential care homes

Map Transparency: 75 %

 Residential care homes

Source

Districts

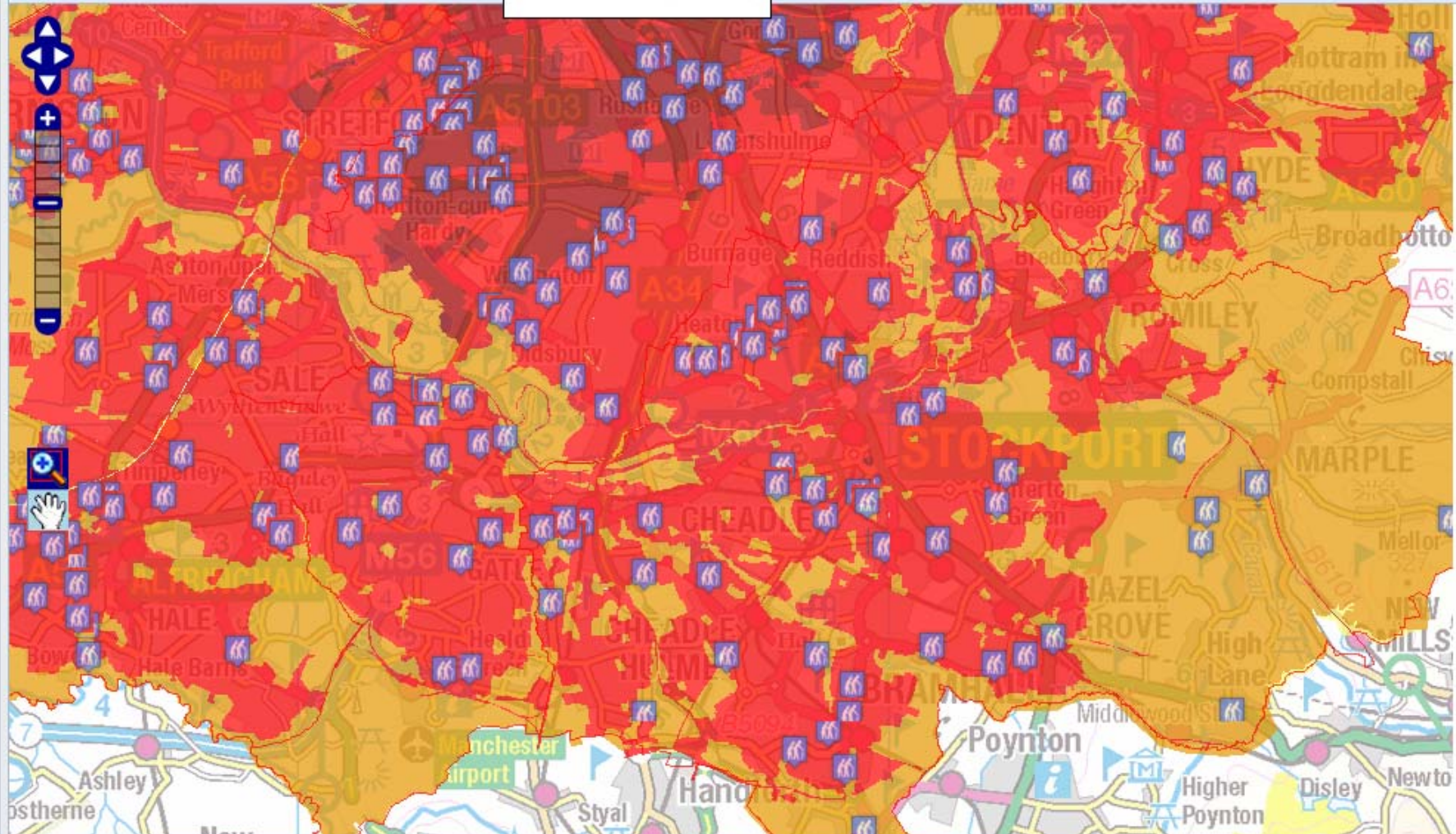
Map Transparency: 75 %

 GM districts

Source

Map

Map Layer Menu



Value of a spatial approach to assessing risks

- Targeting resource allocation
- Encouraging the development of cross-sector collaborative responses
- Communicating risks to the public
- Informing emergency planning
- Supporting long term spatial planning

Progressing climate change adaptation



UK Climate Change Risk Assessment

- Priorities for action within next 5 years:
 - Flood and coastal erosion risk management
 - Specific aspects of natural ecosystems
 - Management of water resources, esp. in areas with increasing water scarcity
 - Overheating of buildings and other infrastructure in the urban environment
 - Risks to health
 - Economic opportunities, esp. to develop adaptation products and services



Source: UK CCRA 2012

Greater Manchester's adaptation objectives

- ***Actions to safeguard future prosperity.*** Adaptation responses can lower perceived climate-related risks associated with inward investment in the area.
- ***Enhancing resilience of critical infrastructure networks.*** Over 60% of water storage and treatment plants are located in Flood Zone 2 (1% annual probability of flooding).
- ***Protecting the most vulnerable communities.*** Areas at risk of surface water flooding in Greater Manchester coincide with communities who are least able to prepare for and deal with floods.

Greater Manchester's key adaptation opportunities

1. Frameworks for natural process (watershed/airshed) broadly link to GM's administrative geography
2. Partnership working with the University sector is gaining strength
3. Conurbation scale governance frameworks are emerging (GMCA and AGMA)
4. Reference to action on adaptation through the GM Strategy and GM Climate Strategy

Acknowledgements

- The EcoCities and GRaBS teams, in particular:
 - Gina Cavan
 - Angela Connelly
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